# Daily Number Theory Drill #5

*Sheet by David Akinyele-Aje*

| Section | Topic | Questions | Target time |
|---|---|---|---|
| A | Rapid Recognition | 10 | 2 min 30 sec |
| B | Manipulation Drills | 10 | 8 min |
| C | Substitution & Structure | 8 | 10 min |
| D | Challenge | 5 | 15 min |

*New toolkit today: Bounding Diophantine Equations · Inequality Logic · Factor Searches.*

*Attempt each question before checking answers. Time yourself. No calculators.*

## Section A Rapid Recognition (≤15 s each)

*Last digits, primes, and basic properties. Pure recognition.*

**1.** Find the last digit of $3^{2025}$.

**2.** Factorise 1001 completely into primes.

**3.** Find $\gcd(42, 105)$.

**4.** Find the number of positive divisors of 120.

**5.** Evaluate $5^{10} \pmod 6$.

**6.** Find the smallest positive integer $x$ such that $2x \equiv 1 \pmod 5$.

**7.** Find the smallest integer $n > 1$ such that $n \equiv 1 \pmod 3$ and $n \equiv 1 \pmod 4$.

**8.** Write the prime factorisation of 360.

**9.** Calculate $7^3 \pmod{10}$.

**10.** Find the largest prime factor of 9999.

## Section B Manipulation Drills (≤50 s each)

*Modular arithmetic, exponentiation, divisibility rules. Look for patterns.*

**1.** The greatest power of 7 which is a factor of 50! is $7^k$. What is $k$? *(after SMC 2023 Q12)*

**2.** Find the smallest positive integer $n$ such that $15n$ is a perfect cube.

**3.** Find the remainder when $1! + 2! + 3! + \cdots + 100!$ is divided by 12.

**4.** Which is larger, $3^{40}$ or $4^{30}$?

**5.** Find the smallest positive integer $x$ such that $3x \equiv 4 \pmod 7$.

**6.** Find the last two digits of $7^{2022}$.

**7.** How many positive factors of 1000 are perfect squares?

**8.** Find all integers $x$ such that $0 \leq x < 8$ and $x^2 \equiv 1 \pmod 8$.

**9.** Given $2^x 3^y = 12^3$, find $x + y$.

**10.** Find all pairs of positive integers $(x, y)$ such that $xy = x + y + 3$.

## Section C Substitution & Structure (≤90 s each)

*Synoptic links: use algebraic identities and combinatorial counting to solve these number theory problems.*

**1.** The expression $\frac{n^2+3n+5}{n+1}$ takes integer values for certain positive integer values of $n$. What is the sum of all such positive integer values of $n$? *(after SMC 2023 Q19)*

**2.** How many of the numbers 6, 7, 8, 9, 10 are factors of the sum $2^{2024} + 2^{2023} + 2^{2022}$? *(after SMC 2023 Q10)*

**3.** Find all pairs of positive integers $(x, y)$ such that $x^2 - y^2 = 17$.

**4.** How many ordered pairs of positive integers $(a, b)$ satisfy $\text{lcm}(a, b) = 300$?

**5.** How many positive integers $n \leq 100$ have exactly 3 positive divisors?

**6.** Find the sum of the digits of the largest integer $n$ such that $n^3 + 100$ is a multiple of $n + 10$.

**7.** Find all triples of positive integers $(x, y, z)$ such that $x \leq y \leq z$ and $xyz = x + y + z$. *(after SMC 2022 Q21)*

**8.** Find the sum of all integer values of $n$ for which $n^2 + 4n + 3$ is a perfect square.

## Section D Challenge (TMUA / SMC difficulty)

*Insight over computation. Each question has a short, elegant solution — find it.*

**1.** Find all prime numbers $p$ such that $p^2 + 2$ is also a prime number.

**2.** Find all pairs of positive integers $(x, y)$ such that $x^2 y = x^2 + 3y + 2$. *(after BMO1 1995 Q1)*

**3.** How many pairs of positive integers $(x, y)$ with $x \leq y$ satisfy $\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$? *(after SMC 2021 Q24)*

**4.** Find the remainder when $1^{2025} + 2^{2025} + \cdots + 10^{2025}$ is divided by 11.

**5.** Find the sum of all integers $n$ for which $n^4 + 4$ is a prime number.

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*"Number theory is the queen of mathematics." — Carl Friedrich Gauss*

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Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths